

PRACTICE EXERCISES:

Solve the following combinatorial problems:

1. A student is shopping for a new computer. He is deciding among 3 desktop computers and 4 laptop computers. What is the total number of computer options?
 - ***A student can have 7 computer options. (3 desktop + 4 laptop computers = 7 total options.)***
2. If two dice are rolled, one red and one blue, how many outcomes are possible?
 - ***Red = 6 possible outcomes x Blue = 6 possible outcomes. So $6 \times 6 = 36$ outcomes.***
3. A man has eight shirts, four pants, and five pairs of shoes. How many different outfits are possible?
 - ***A man can wear 160 different outfits. The number of shirts is 8 x The number of pants is 4 x The number of pair of shoes is 5. So $8 \times 4 \times 5$ is equal to 160.***
4. How many different license plates are there if a standard plate consists of three letters followed by three digits?
 - **For each letter, there are 26 possible choices**
 - **For each number there are 10 possible choices**
 - **To calculate it, three letters = $26 \times 26 \times 26$, and Three digits = $10 \times 10 \times 10$, We could have $17,576 \times 1000 = 17,576,000$ different license plates possible.**